

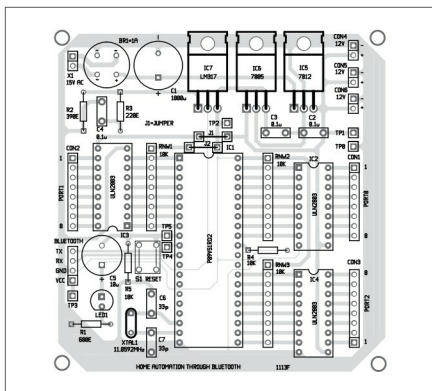


Fig. 9: Component layout for the PCB in Fig. 7

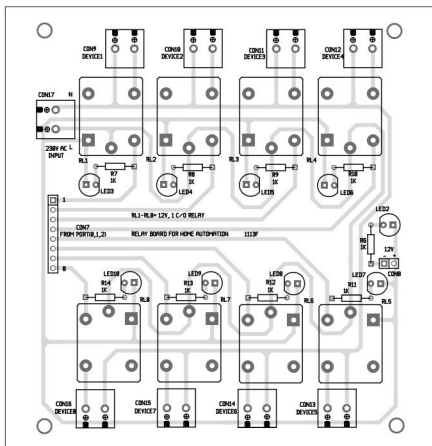


Fig. 10: Component layout for the PCB in Fig. 8

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

professional software developers, such as hobbyists and students.

The application sends ASCII data using the PC's virtual serial port. The method shown is the same for VB2008 as well as VB2012 Express Editions, and also runs on .NET versions. To communicate with the controlling unit using Visual Basic, the SerialPort component must be utilised to allow serial data transfer via a virtual serial port. SerialPort is a component shipped with VB2008 Express Edition and must be loaded using the Toolbox/Components menu. The program has a pull-down menu for serial port selection. There are also the connect, disconnect and exit buttons, to initiate their respective operations. There are three different frames, one for each port, which you can use to switch devices on or off, individually or simultaneously.

Construction and testing

A single-side PCB (in actual size) for the 'Home Automation through Bluetooth' application is shown in Fig. 7, and its component layout is shown in Fig. 9. The PCB for the relay board is shown in Fig. 8, and its component layout is shown in Fig. 10. Assemble the circuit on the recommended PCBs to save assembly time and minimise errors. Suitable connectors are provided on both the PCBs to connect them to each other and to load.

To test the circuit for proper functioning, verify the 12V, 5V and 3.3V power supplies at TP1, TP2 and TP3, respectively, with respect to TP0. The reset to the system can be checked at TP5. The transmitted data by the Bluetooth module can be observed at TP4, using an oscilloscope. ●

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